Solution Set: Central Tendency & Spread Practice Assignment

Task 1: Mean

Dataset:
12, 18, 20, 22, 30, 12, 18, 20, 22, 30, 12, 18, 20, 22, 30
Step 1: Add all the values:
→ 12 + 18 + 20 + 22 + 30 = 102 (one set)
→ 102 × 3 = **306**

Step 2: Count the data points:
→ 15 numbers total

Step 3: Divide to find the mean:
→ Mean = 306 ÷ 15 = **20.4**

Interpretation:
The mean tells us the average value in this dataset. It gives a sense of the overall "center" of the data.

Task 2: Median

Original Dataset (15 values):
3, 7, 8, 5, 12, 3, 7, 8, 5, 12, 3, 7, 8, 5, 12
Step 1: Sort the data:
3, 3, 3, 5, 5, 5, 7, 7, 7, 8, 8, 8, 12, 12, 12

Step 2: Middle value:
→ 15 values → Median = 8th value = **7**

With additional value 15 (16 values):
New dataset:
3, 3, 3, 5, 5, 5, 7, 7, 7, 8, 8, 8, 12, 12, 12, 15

→ Median = average of 8th & 9th values → (7 + 7) ÷ 2 = **7**

Interpretation:
With a symmetric dataset and an added higher value, the median remains the same because most values are concentrated in the center.

Task 3: Mode

Dataset:
9, 15, 9, 12, 9, 18, 15, 18, 15, 9, 12, 9, 18, 15, 18, 15, 9, 12, 9, 18, 15, 18

Frequency Count:

- 9 → 7 times
- 15 → 6 times
- 18 → 6 times
- 12 → 3 times

Mode:
→ **9**

Explanation:
The mode is the value that occurs most frequently. In this case, 9 appears most often, making it the mode.

Task 4: Range

Dataset:
22, 25, 18, 30, 28 — repeated 3 times (values repeated don't affect range)

Min value: 18
Max value: 30
Range = Max − Min = 30 − 18 = 12

Interpretation:
The range shows the overall spread of the data from smallest to largest value.

Task 5: Variance & Standard Deviation

Dataset:
10, 12, 14, 16, 18

Step 1: Mean = (10 + 12 + 14 + 16 + 18) ÷ 5 = **14**

Step 2: Squared differences from the mean:

- $(10 - 14)^2 = 16$
- $(12 - 14)^2 = 4$

- $(14 - 14)^2 = 0$
- $(16 - 14)^2 = 4$
- $(18 - 14)^2 = 16$

Step 3: Variance = (16 + 4 + 0 + 4 + 16) ÷ 5 = **8**

Step 4: Standard Deviation = √8 ≈ **2.83**

Interpretation:
The variance and standard deviation help us understand how much the values differ from the average. A smaller standard deviation means the values are closer to the mean.

Task 6: Applying All Concepts

Dataset:
5, 10, 10, 15, 20, 25, 30

Mean:
→ (5 + 10 + 10 + 15 + 20 + 25 + 30) ÷ 7 = 115 ÷ 7 = **16.43**

Median:
→ Middle value = 4th value = **15**

Mode:
→ 10 appears twice → **Mode = 10**

Range:
→ 30 − 5 = **25**

Variance:
Mean = 16.43
→ Squared differences:
$(5-16.43)^2 \approx 130.18$
$(10-16.43)^2 \approx 41.35$
$(10-16.43)^2 \approx 41.35$
$(15-16.43)^2 \approx 2.04$
$(20-16.43)^2 \approx 12.68$
$(25-16.43)^2 \approx 73.52$
$(30-16.43)^2 \approx 185.08$
→ Total = 486.2
→ Variance = 486.2 ÷ 7 ≈ **69.46**

Standard Deviation:
$\sqrt{69.46} \approx$ **8.33**

Best Measure of Center:
Median = 15 is a better representation here because the data has some spread, and median isn't affected by outliers as much as the mean.

Task 7: Reflection

Measures of Central Tendency (Mean, Median, Mode):
These describe where the "center" of the data lies. For example, the median is often used for income data because it's not thrown off by very high or low incomes.

Measures of Spread (Range, Variance, Standard Deviation):
These tell us how spread out the values are. For example, standard deviation is key in finance to measure how volatile a stock is.

Why It Matters:
Understanding both the center and the spread of data gives a full picture. You can't just look at the average—you need to know how much variation there is to make good decisions based on data.